

CSE 6740: Computational Data Analysis

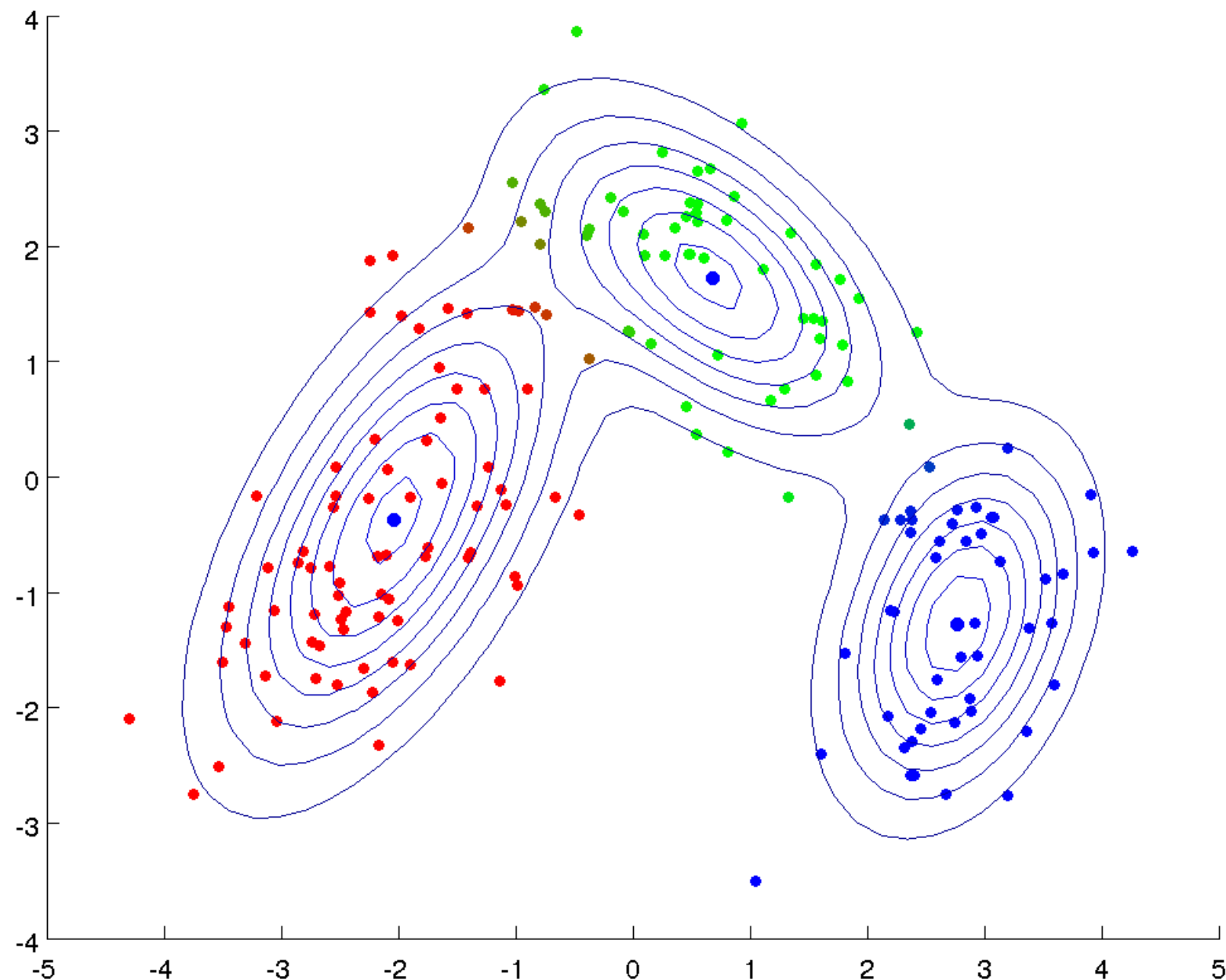
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Gaussian Mixture Model and EM

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Wine dataset

- First run PCA to reduce the dimension to 2
- Clear cluster structure
- Can we fit 3 Gaussians?



Parametric vs. nonparametric

- Data $x \in R^d$ with **fixed** dimension d
- Given m training data points $\{x^1, x^2 \dots, x^m\}$
- Partition n bin in each dimension

Aspects	Gaussian	Histogram	KDE
Flexible	Not	Yes	Yes
Assumption	Strong	Not	Not
Parameter number	Fixed	Increase with n	Increase with m
Memory requirement	$d+d^2$	n^d	md
Training computation	Closed form	Binning and Counting	Nothing
Test computation	Plug in formula	Find the bin	Evaluate m functions
Statistical guarantee	Only Gaussian case	Arbitrary (worse)	Arbitrary (better)

- A density model $p(X)$ may be multi-modal: model it as a mixture of uni-modal distributions (e.g. Gaussians)

$$\mathcal{N}(X|\mu_k, \Sigma_k) := \frac{1}{|\Sigma|^{1/2} (2\pi)^{d/2}} \exp\left(-\frac{1}{2} (X - \mu)^T \Sigma^{-1} (X - \mu)\right)$$

- Consider a mixture of K Gaussians

- $p(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$

mixing
proportion

mixture
Component

- Parametric or nonparametric?

- Learn $\pi_k \in (0,1), \mu_k, \Sigma_k$;

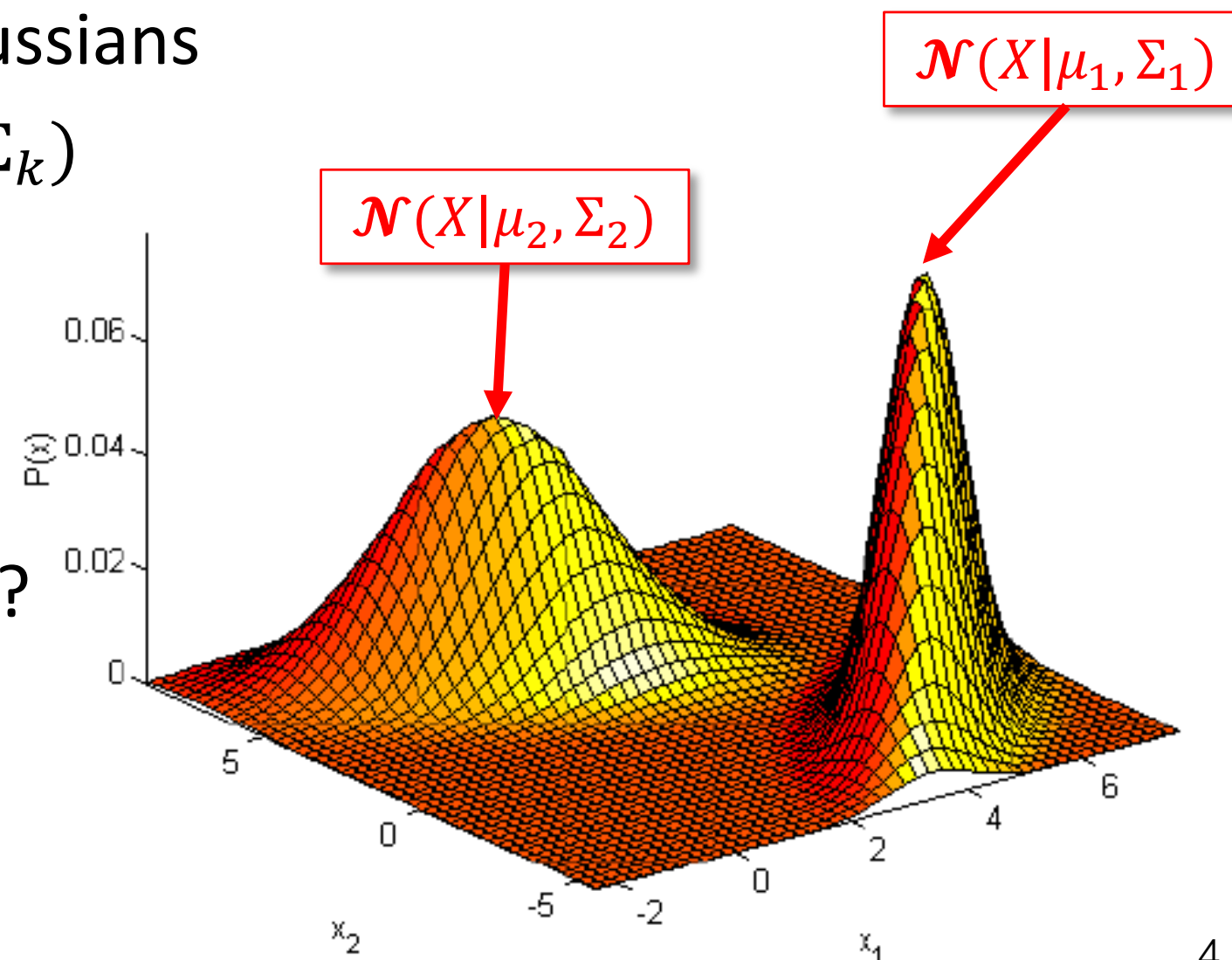


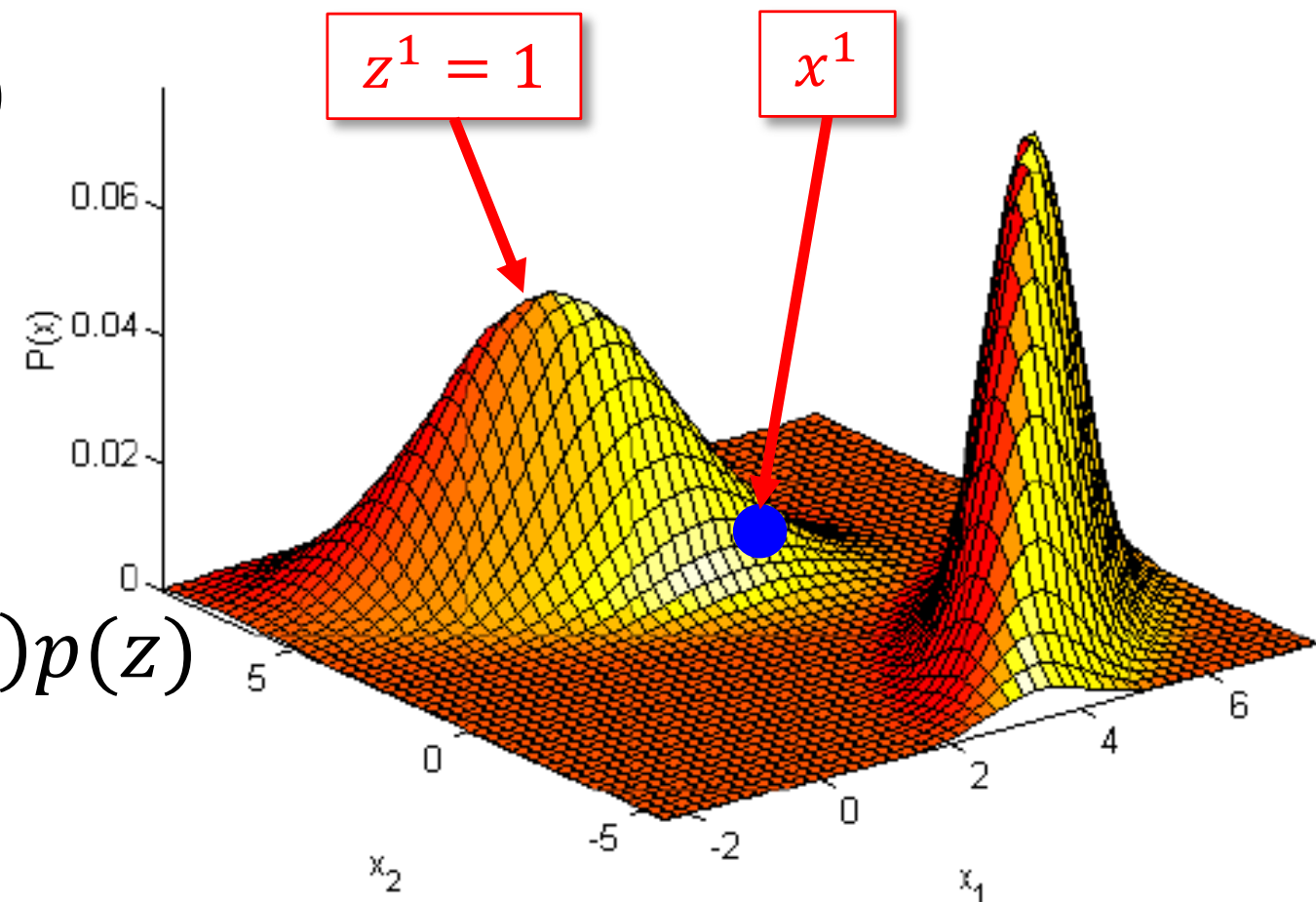
Image a generative process for data points

- For each data point x^i :
 - Randomly choose a mixture component, $z^i = \{1, 2, \dots, K\}$, with probability π_{z^i}
 - Then sample the actual value of x^i from a Gaussian distribution $\mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i})$

- Joint distribution over $p(x, z)$
 $p(x, z) = \pi_z \mathcal{N}(x | \mu_z, \Sigma_z)$

- Marginal distribution $p(x)$

$$p(x) = \sum_{z=1}^K p(x, z) = \sum_{z=1}^K p(x|z)p(z)$$



- How to learn?
- Maximum likelihood learning (let $\theta = (\pi_k, \mu_k, \Sigma_k), k = 1 \dots K$)

$$\theta^* = \operatorname{argmax} l(\theta; D) = \log \prod_{i=1}^m p(x^i)$$

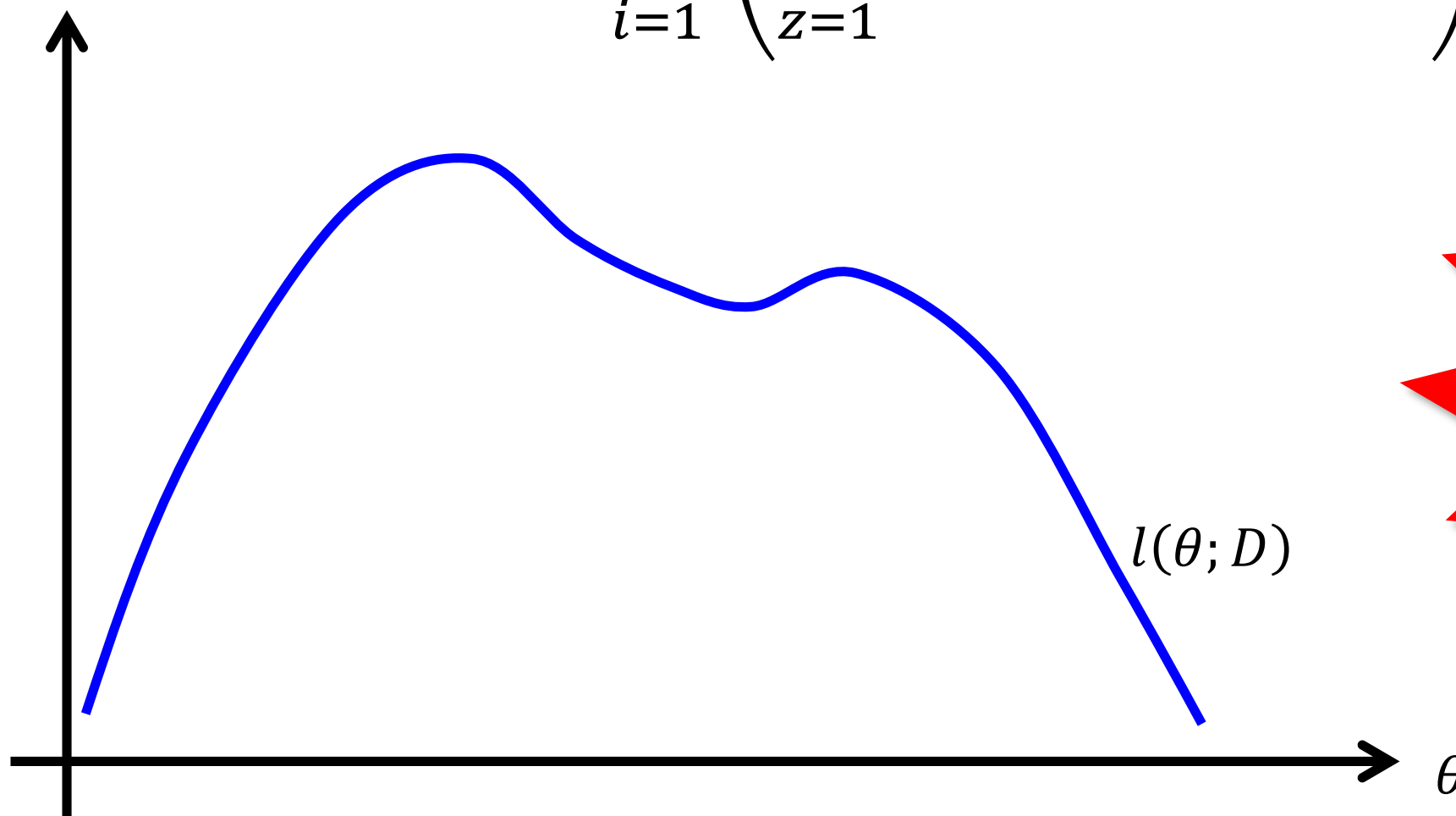
- Use our generative process

$$\begin{aligned} l(\theta; D) &= \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i, z^i | \theta) \right) \\ &= \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right) \end{aligned}$$

Why is learning hard?

- With latent variables z , likelihood of the data becomes

$$l(\theta; D) = \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right)$$
$$= \log \prod_{i=1}^m \left(\sum_{z=1}^K \pi_{z^i} \mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i}) \right)$$



**Nonconvex
Difficult!**

$$\pi_N = \mathbf{1}$$

$$\log(\alpha_1 \cdot 1 + \alpha_2 \cdot 0) \approx 1 \cdot \log \alpha_1 + 0 \cdot \log \alpha_2$$

When z is given

- Let us introduce a binary variable τ_k^i to indicate whether $z^i = k$

$$l(\theta; D) = \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right)$$

$$= \log \prod_{i=1}^m \left(\sum_{z=1}^K \pi_{z^i} \mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i}) \right)$$

$$= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \log \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)$$

$$\begin{aligned} &= \sum_{i=1}^m \log \sum_{z=1}^K \pi_{z^i} \mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i}) \\ &= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \log \pi_k \end{aligned}$$

$$= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \left[\log \pi_k - (x^i - \mu_k)^\top \Sigma_k^{-1} (x^i - \mu_k) + \log |\Sigma_k| + c \right]$$

When z is given

- Let us introduce a binary variable τ_k^i to indicate whether $z^i = k$

$$l(\theta; D) = \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \left[\log \pi_k - (x^i - \mu_k)^\top \Sigma_k (x^i - \mu_k) + \log \Sigma_k + c \right]$$

- It is now easy to show that, the maximum likelihood estimation of the parameters are

$$\begin{aligned} \pi_k &= \frac{\sum_i \tau_k^i}{m}, & \mu_k &= \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i} \\ \Sigma_k &= \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^\top}{\sum_i \tau_k^i} \\ &\quad (k = 1 \dots K) \end{aligned}$$

When z is not given

- We can guess which Gaussian x^i comes from by computing the posterior probability

$$p(z^i = k | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})} \quad (k = 1 \dots K, i = 1 \dots m)$$

- Bayes rule

likelihood Prior

Prior: $p(z) = \pi_z$
Likelihood: $p(x|z) = \mathcal{N}(x|\mu_z, \Sigma_z)$

$$P(z|x) = \frac{P(x|z)P(z)}{P(x)} = \frac{P(x, z)}{\sum_{z'} P(x, z')}$$

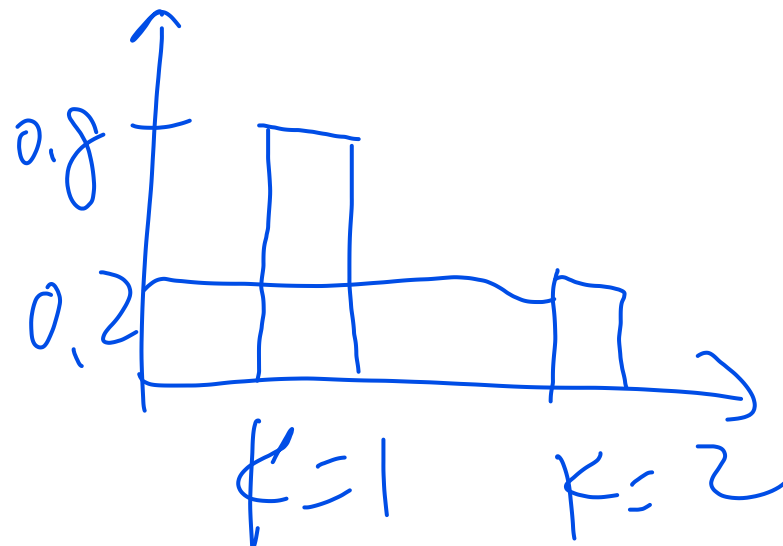
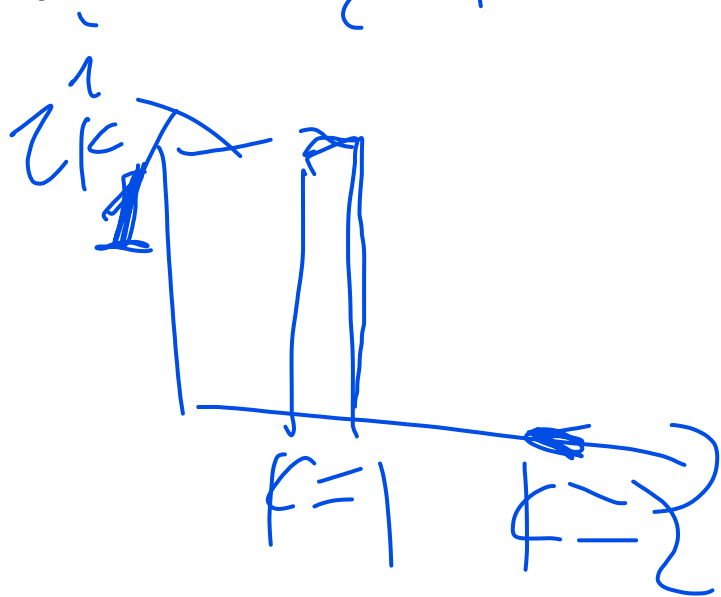
posterior normalization constant

When z is not given

- We can guess which Gaussian x^i comes from by computing the posterior probability

$$p(z^i = k | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})} \quad (k = 1 \dots K, i = 1 \dots m)$$

- We are going to pretend $p(z_k^i = 1 | D, \mu, \Sigma)$ as τ_k^i which should be binary – but now is regarded as “soft” assigning x^i to k -th component. z_k^i



- Associate each data and each component with a τ_k^i
- Initialize (π_k, μ_k, Σ_k) , $k = 1 \dots K$
- Iterate the following two steps till convergence:
 - **Expectation step (E-step)**: update τ_k^i given current (π_k, μ_k, Σ_k)

$$\tau_k^i = p(z_k^i = 1 | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})}$$

$(k = 1 \dots K, i = 1 \dots m)$

- **Maximization step (M-step)**: update (π_k, μ_k, Σ_k) given τ_k^i

$$\pi_k = \frac{\sum_i \tau_k^i}{m}, \quad \mu_k = \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i}$$
$$\Sigma_k = \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^T}{\sum_i \tau_k^i}$$

$(k = 1 \dots K)$

- We intend to learn the parameters that maximizes the log-likelihood of the data

$$l(\theta; D) = \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i, z^i | \theta) \right)$$



Nonconvex
Difficult!

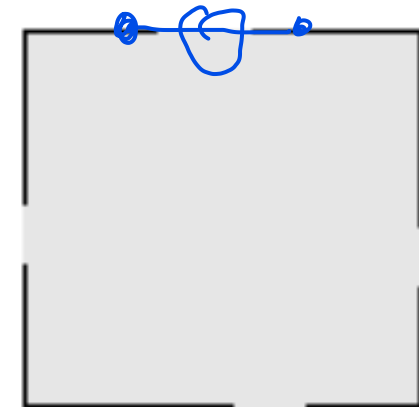
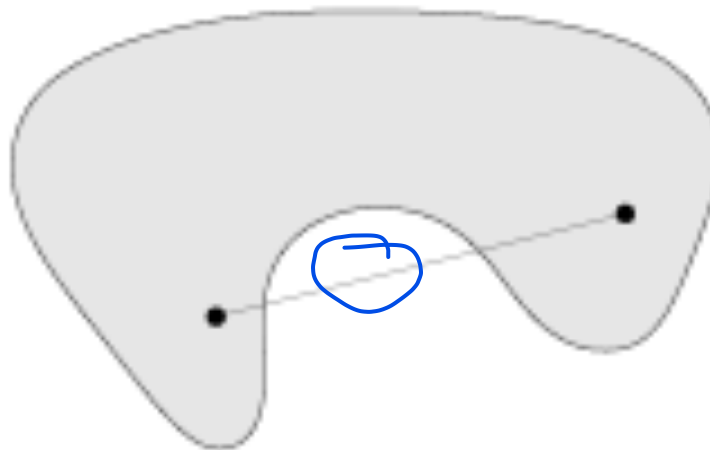
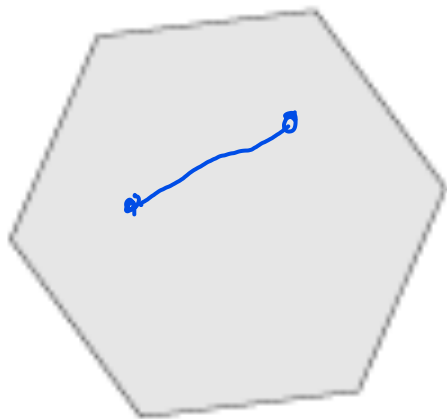
- Expectation step (E-step): What do we take expectation over?

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[\log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

- Maximization step (M-step): how to maximize?

$$\theta^{t+1} = \operatorname{argmax}_{\theta} f(\theta)$$

- Definition: A set A is convex, if for every $0 \leq \alpha \leq 1$ it satisfies
 - $\forall x, y \in A \rightarrow \alpha x + (1 - \alpha)y \in A$ $\alpha \in [0, 1]$
- The line segment between any two points is also in the set.
- Examples of convex and non-convex sets



Convex Functions

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & g(x) \leq 1 \end{array} \quad x \in \mathbb{R}^n$$

- Definition: A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is **convex** if the domain $\text{dom} f$ is a convex set and if for all $x, y \in \text{dom} f$, and $0 \leq \theta \leq 1$, we have

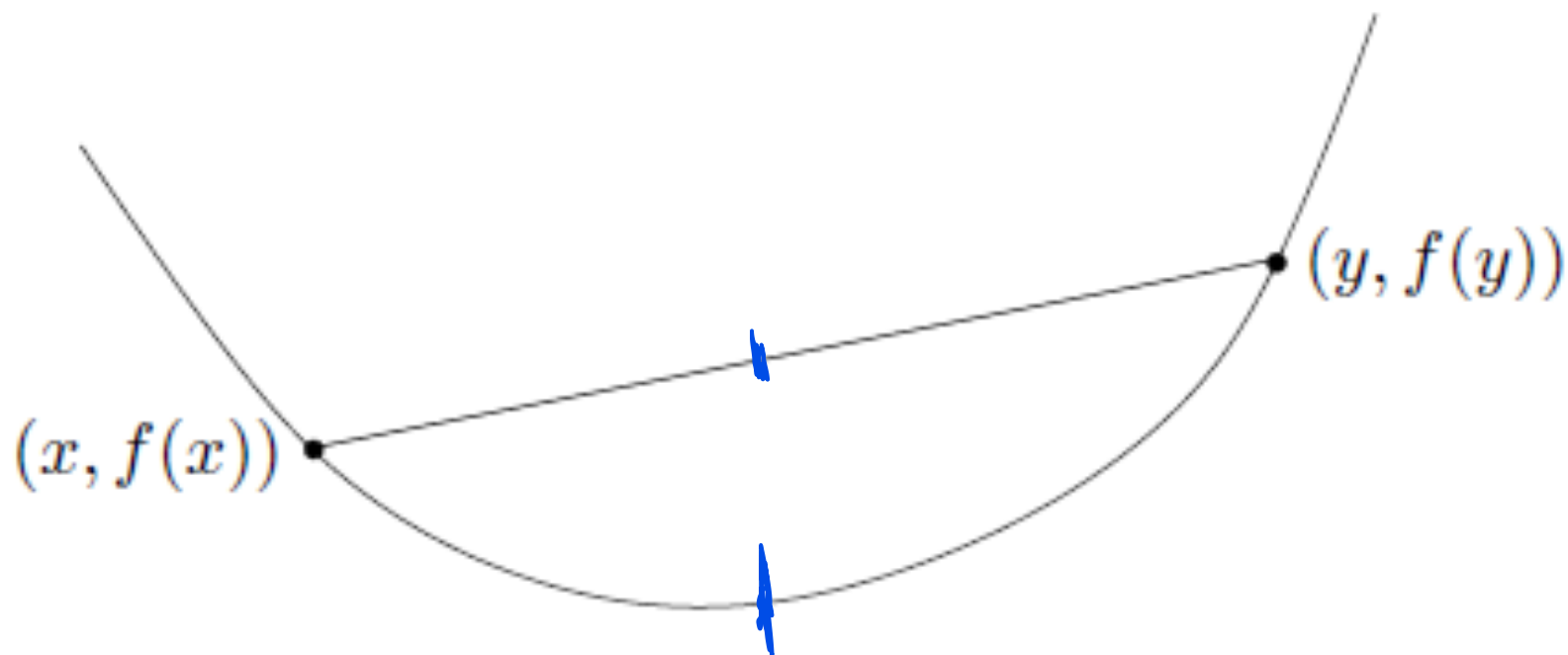
$$f(x) = x$$

$$f(x) = x$$

$$\underline{f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)}$$

- Geometrically, the line segment between $(x, f(x))$ and $(y, f(y))$ lies **above** the graph of f

x



- Definition: A function $f: R^n \rightarrow R$ is **concave** if the domain $\mathbf{dom} f$ is a convex set and if for all $x, y \in \mathbf{dom} f$, and $0 \leq \theta \leq 1$, we have

$$f\left(\sum_{i=1}^m \theta_i x_i\right) \geq \sum_{i=1}^m f(x_i) \cdot \theta_i$$

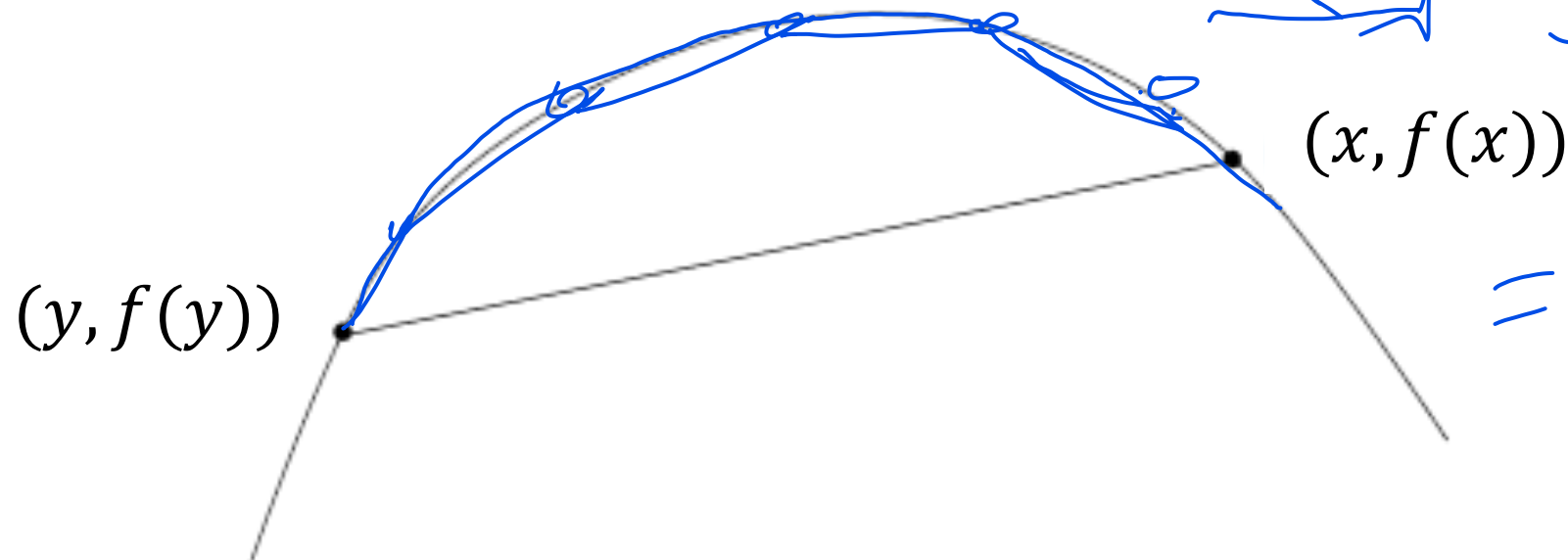
$$f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$$

$$\sum_{i=1}^m \theta_i = 1$$

$$\theta_i \geq 0$$

$$f(E[X]) \geq E[f(X)]$$

- Geometrically, the line segment between $(x, f(x))$ and $(y, f(y))$ lies **below** the graph of f



$$E[X] = \int x p(x) dx$$

$$= \sum_{i=1}^m f(x_i) p(x_i)$$

- We intend to learn the parameters that maximizes the log-likelihood of the data

$$l(\theta; D) = \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i, z^i | \theta) \right)$$



Nonconvex
Difficult!

- Expectation step (E-step): What do we take expectation over?

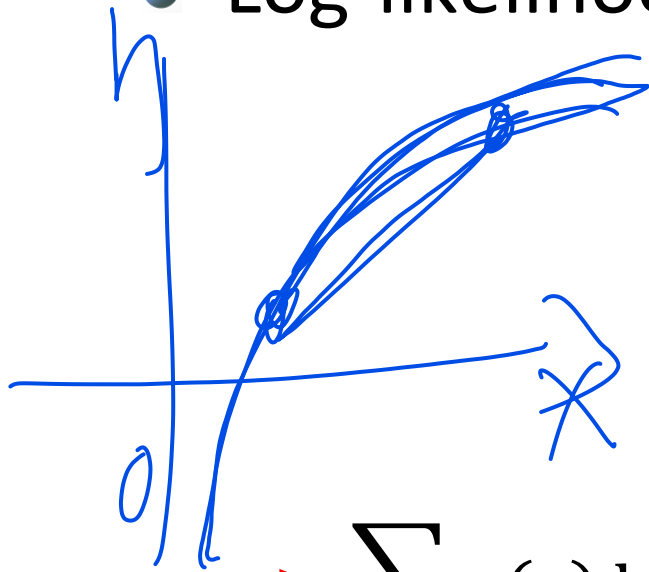
$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[\log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

- Maximization step (M-step): how to maximize?

$$\theta^{t+1} = \operatorname{argmax}_{\theta} f(\theta)$$

Lower bound of log-likelihood

- Log-likelihood $l(x; \theta) = \log \sum_z p(x, z | \theta)$



$$= \log \sum_z q(z) \frac{p(x, z | \theta)}{q(z)} \text{ (arbitrary } q(z) \text{)}$$

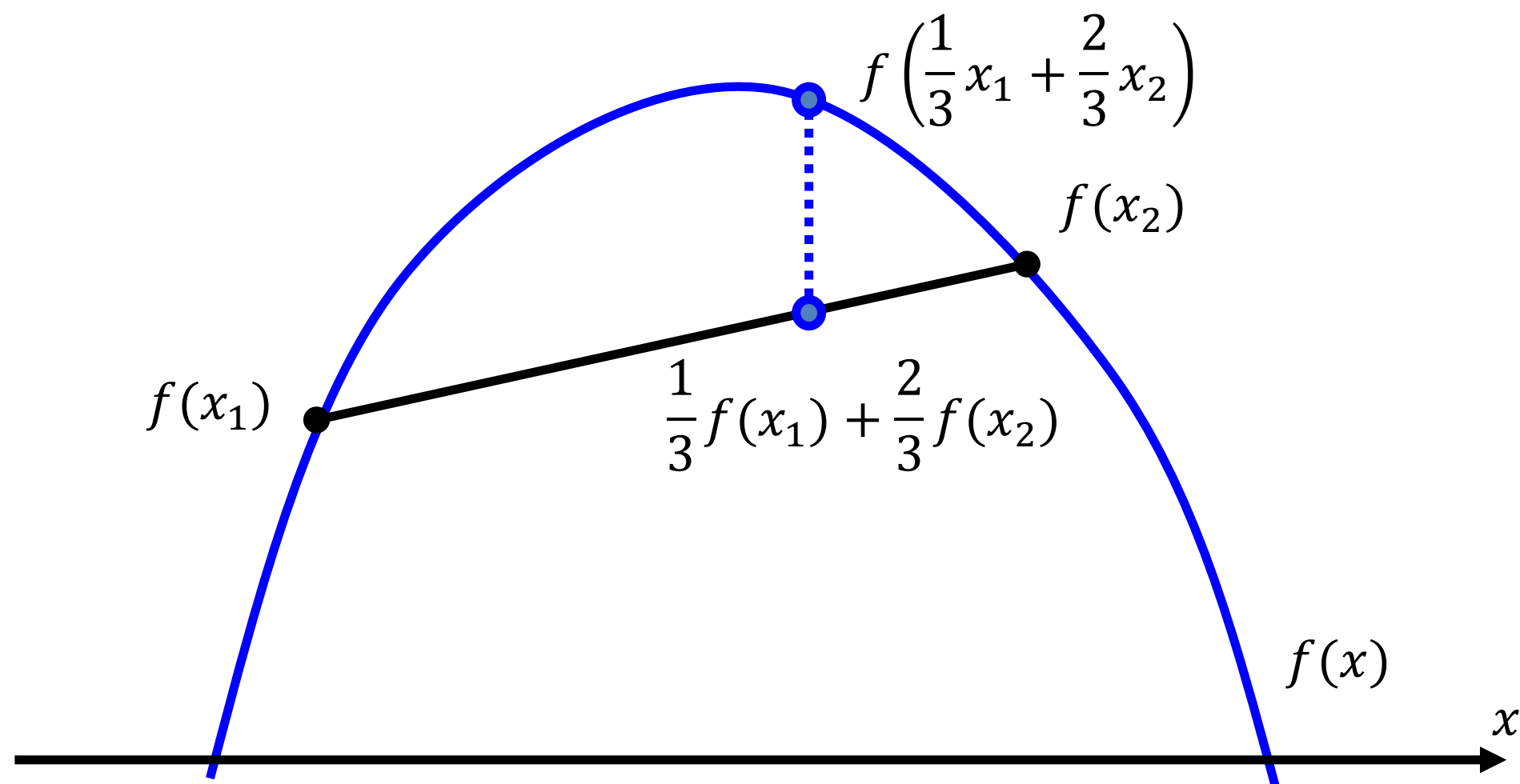
$$\geq \sum_z q(z) \log \frac{p(x, z | \theta)}{q(z)} \text{ (Jensen's inequality } f\left(\sum_i \alpha_i x_i\right) \geq \sum_i \alpha_i f(x_i))$$

$$= \sum_z q(z) \log p(x, z | \theta) - \sum_z q(z) \log q(z)$$

$$= E_{q(z)}[\log p(x, z | \theta)] + H_{q(z)}$$

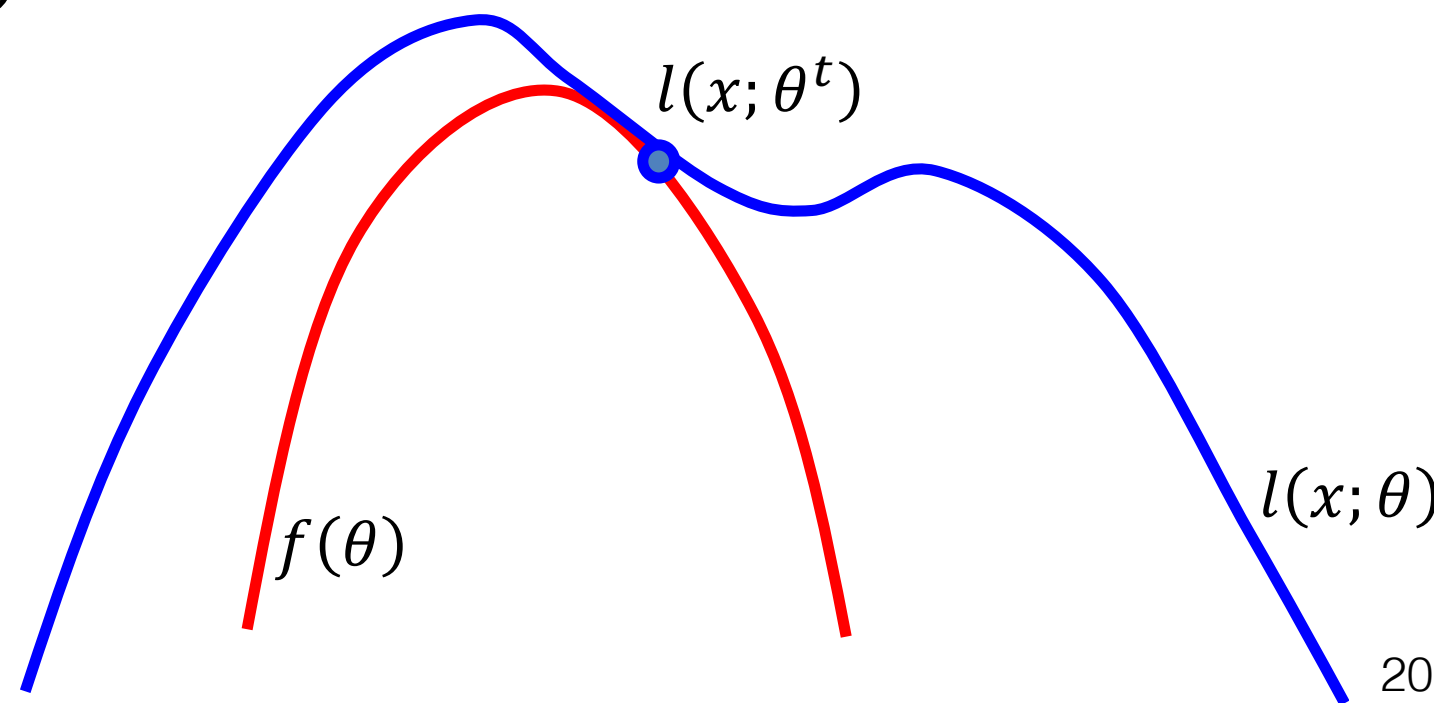
Jensen's inequality

- For concave function $f(x)$
 - $f(\sum_i \alpha_i x_i) \geq \sum_i \alpha_i f(x_i)$, where $\sum_i \alpha_i = 1, \alpha_i \geq 0$
- Most general case: If x is a random variable, and f is concave,
$$f(\mathbf{E}x) \geq \mathbf{E}f(x)$$

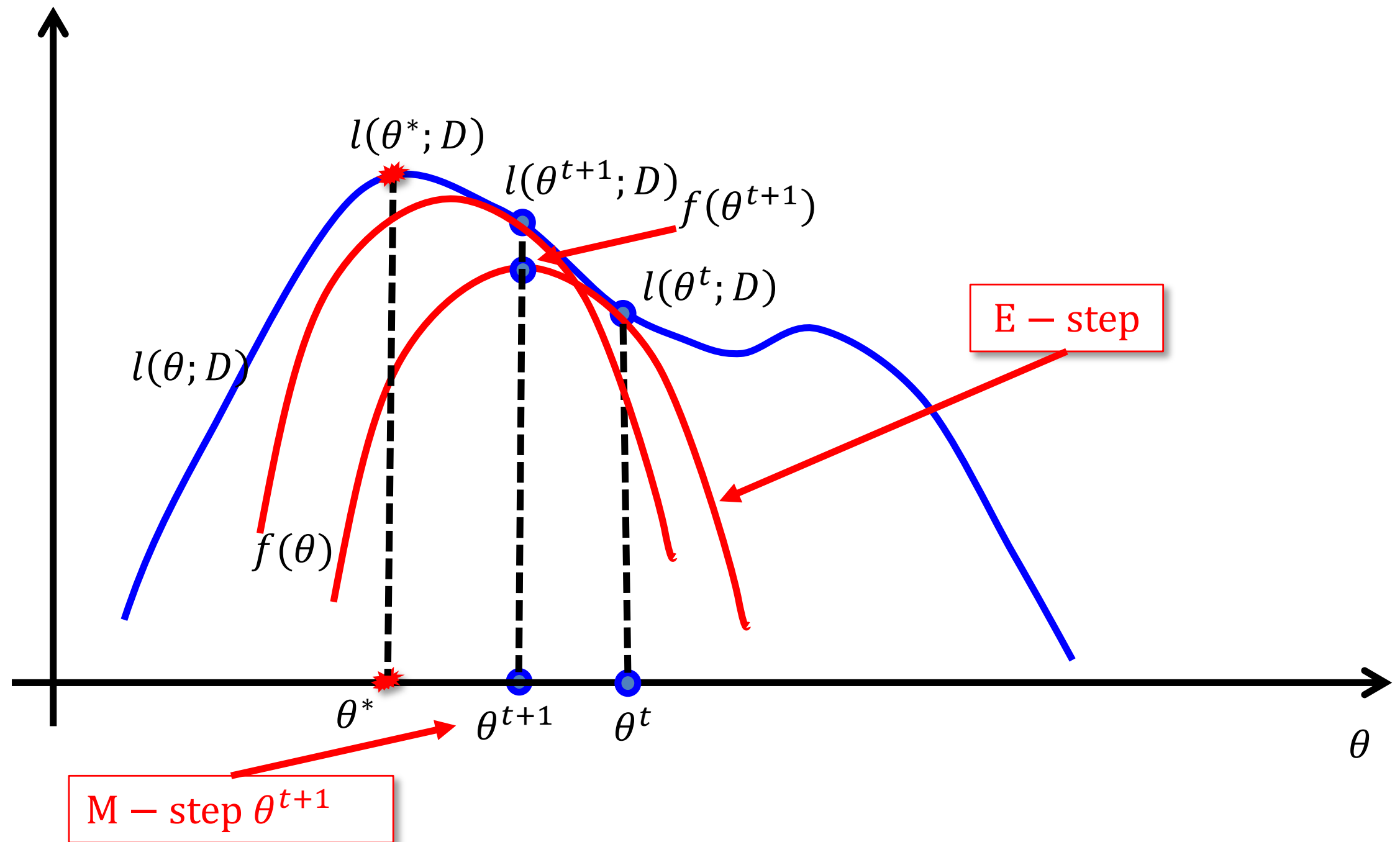


What attains equality?

- $q(z) = p(z|x, \theta^t)$: posterior of z given x attains the equality at θ^t
- Let $F(q, \theta) = \sum_z q(z|x) \log \frac{p(x, z|\theta)}{q(z|x)} \leq l(x; \theta) = \log \sum_z p(x, z|\theta)$
- $F(p(z|x, \theta^t), \theta^t) = \sum_z p(z|x, \theta^t) \log \frac{p(x, z|\theta^t)}{p(z|x, \theta^t)}$
- $= \sum_z p(z|x, \theta^t) \log p(x|\theta^t)$
- $= \log p(x|\theta^t)$
- $= \log \sum_z p(x, z|\theta^t)$



EM graphically



- Why is $f(\theta)$ a lower bound? ✓

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[\log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

- Why $q(z^1, z^2, \dots, z^m) = \prod_{i=1}^m p(z^i | x^i, \theta^t)$ ✓

- Why will EM converge? ✓

Expectation step (E-step): What do we take expectation over?

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[\log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

Maximization step (M-step): how to maximize?

$$\theta^{t+1} = \operatorname{argmax}_{\theta} f(\theta)$$

E-step: what is $q(z^1, z^2, \dots, z^m)$

- $q(z^1, z^2, \dots, z^m)$: posterior distribution of the latent variables

$$q(z^1, z^2, \dots, z^m) = \prod_{i=1}^m p(z^i | x^i, \theta^t)$$

- For each data point x^i , compute $p(z^i = k | x^i)$ for each k

$$\tau_k^i = p(z^i = k | x^i) = \frac{p(z^i = k, x^i)}{\sum_{k'=1..K} p(z^i = k', x^i)}$$

$$= \frac{\pi_k \mathcal{N}(x^i | \mu_k, \Sigma_k)}{\sum_{k'=1..K} \pi_{k'} \mathcal{N}(x^i | \mu_{k'}, \Sigma_{k'})}$$

E-step: compute the expectation

$$\begin{aligned} f(\theta) &= E_{q(z^1, z^2, \dots, z^m)} \left[\log \prod_{i=1}^m p(x^i, z^i | \theta) \right] \\ &= \sum_{i=1}^m E_{p(z^i | x^i, \theta^t)} [\log p(x^i, z^i | \theta)] \\ &= \sum_{i=1}^m E_{p(z^i | x^i, \theta^t)} [\log \pi_{z^i} \mathcal{N}(x^i | \mu_{z^i}, \Sigma_{z^i})] \end{aligned}$$

- Expand log of Gaussian $\log \mathcal{N}(x^i | \mu_{z^i}, \Sigma_{z^i})$

$$\begin{aligned} f(\theta) &= \sum_{i=1}^m E_{p(z^i | x^i, \theta^t)} \left[\log \pi_{z^i} - (x^i - \mu_{z^i})^\top \Sigma_{z^i}^{-1} (x^i - \mu_{z^i}) + \log \Sigma_{z^i} + c \right] \\ &= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \left[\log \pi_k - (x^i - \mu_k)^\top \Sigma_k^{-1} (x^i - \mu_k) + \log \Sigma_k + c \right] \end{aligned}$$

M-step: maximize $f(\theta)$

- $f(\theta) = \sum_{i=1}^m \sum_{k=1}^K \tau_i^k \left[\log \pi_k - (x^i - \mu_k)^\top \Sigma_k (x^i - \mu_k) + \log \Sigma_k + c \right]$

- For instance, we want to find π_k , and $\sum_{i=1}^K \pi_k = 1$

- Form Lagrangian

$$L = \sum_{i=1}^m \sum_{k=1}^K \tau_k^i [\log \pi_k + \text{other terms}] + \lambda (1 - \sum_{i=1}^K \pi_k)$$

$$\frac{1}{\lambda} \sum_{i=1}^m 1 = 1$$

$$\lambda = m$$

- Take partial derivative and set to 0

$$\frac{\partial L}{\partial \pi_k} = \sum_{i=1}^m \frac{\tau_k^i}{\pi_k} - \lambda = 0$$

$$\Rightarrow \pi_k = \frac{1}{\lambda} \sum_{i=1}^m \tau_k^i$$

$$\Rightarrow \lambda = m$$

$$\frac{1}{\lambda} \sum_{i=1}^m \sum_{k=1}^K \tau_k^i = 1$$

- Associate each data and each component with a τ_k^i
- Initialize (π_k, μ_k, Σ_k) , $k = 1 \dots K$
- Iterate the following two steps till convergence:
 - **Expectation step (E-step)**: update τ_k^i given current (π_k, μ_k, Σ_k)

$$\tau_k^i = p(z_k^i = 1 | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})}$$

$(k = 1 \dots K, i = 1 \dots m)$

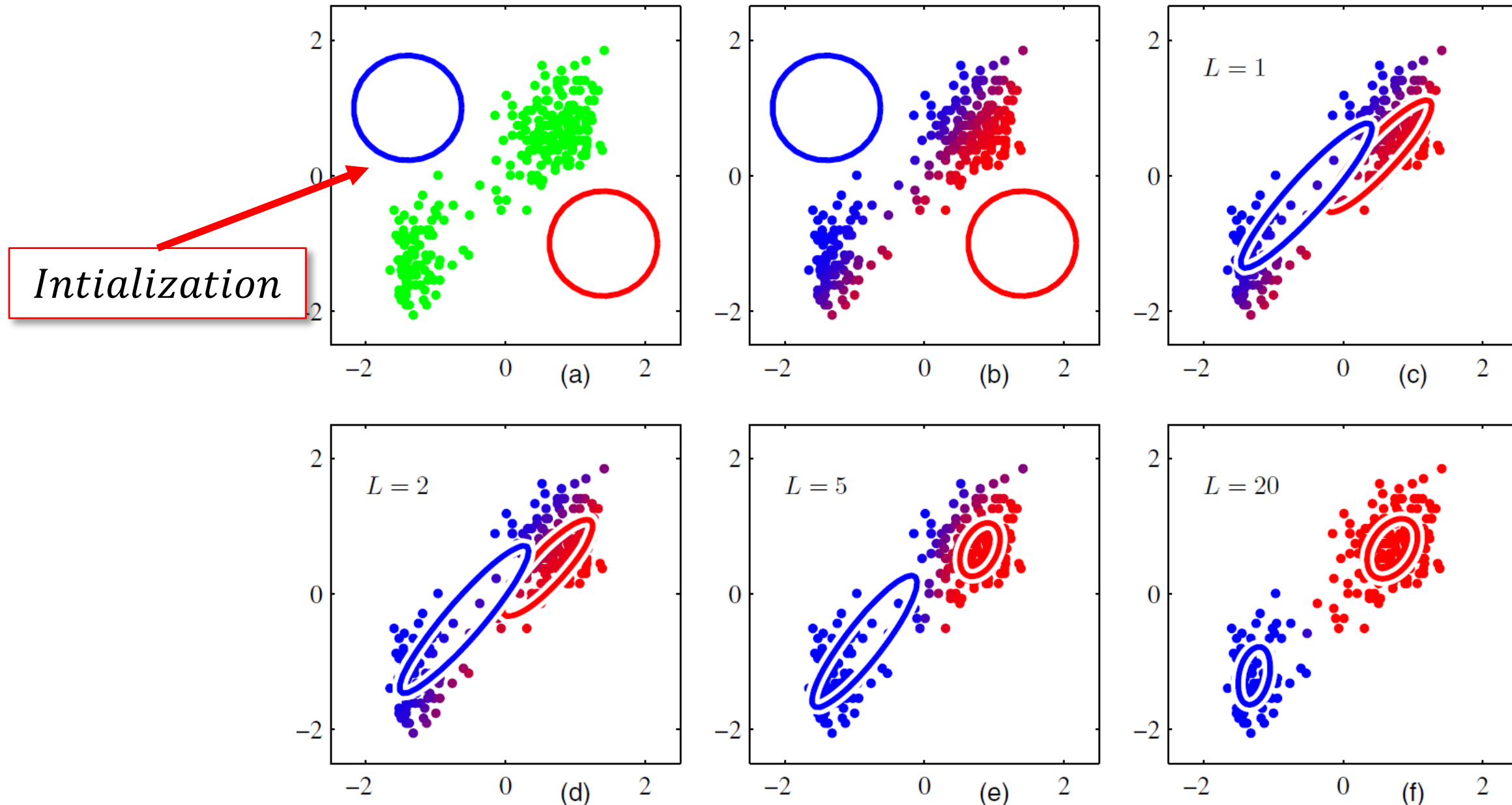
- **Maximization step (M-step)**: update (π_k, μ_k, Σ_k) given τ_k^i

$$\pi_k = \frac{\sum_i \tau_k^i}{m}, \quad \mu_k = \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i}$$
$$\Sigma_k = \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^T}{\sum_i \tau_k^i}$$

$(k = 1 \dots K)$

Expectation-Maximization Iterations

- $k = 1 \text{ or } 2$
- Use τ_1^i as the proportion of red, and τ_2^i proportion of blue
- Draw only one contour for each Gaussian component



EM vs. modified K-means

- The EM algorithm for mixture of Gaussian is like a soft clustering algorithm
 - K-means:
 - “E-step”, we do hard assignment:
 - $z^i = \operatorname{argmax}_k (x^i - \mu_k)^T \Sigma_k^{-1} (x^i - \mu_k)$
 - “M-step”, we update the means and covariance of cluster using maximum likelihood estimate:
 - $\mu_k = \frac{\sum_i \delta(z^i, k) x^i}{\sum_i \delta(z^i, k)}$
 - $\Sigma_k = \frac{\sum_i \delta(z^i, k) (x^i - \mu_k) (x^i - \mu_k)^T}{\sum_i \delta(z^i, k)}$
- $\delta(z^i, k) = 1$ if $z^i = k$; otherwise 0.

General applicability of EM algorithm

- Applicable to other models with latent (or missing) variables
- Example:
 - Expectation maximization applied to a coin toss example
 - Assume you have five observations of coin flips from two coins but you don't know from which coin each of the observations is from
 - The EM algorithm starts by initializing a random prior
 - Then it calculates the expected log probability distribution over the observations, and based on the log probability updates the prior